

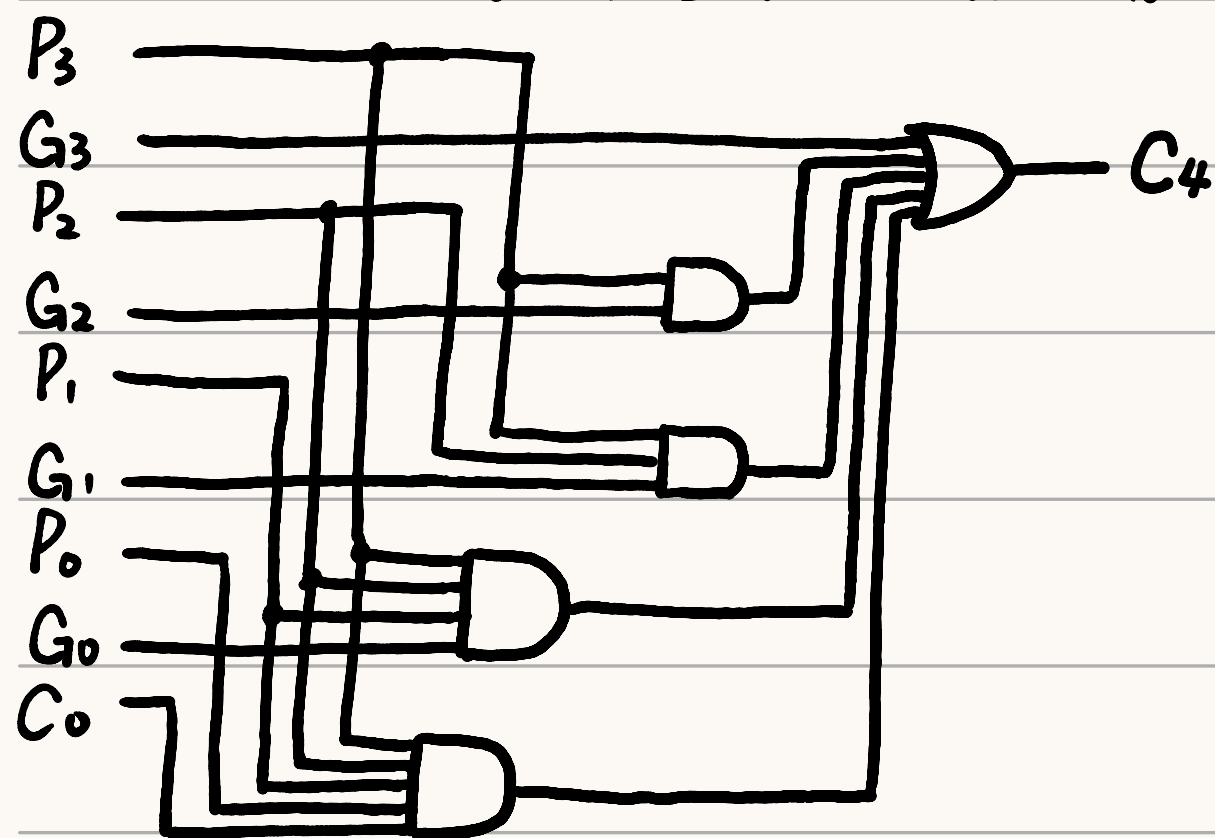
HW4

In slides

$$\begin{aligned}
 4-(1) \quad xy + xz + yz &= xy + x(y+y')z + (x+x')yz \\
 &= xy + xy'z + xy'z + x'yz = xy(1+z) + xy'z + x'yz \\
 &= xy + z(x'y + xy') = (x \oplus y)z + xy
 \end{aligned}$$

$$4-(2) \quad C_4 = G_3 + P_3 C_3$$

$$= G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0$$



4-(3)

E	x	y	z	w	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	D ₈	D ₉	D ₁₀	D ₁₁	D ₁₂	D ₁₃	D ₁₄	D ₁₅
00	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
01	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
00	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
00	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
00	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
01	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
01	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
01	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
00	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
00	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
00	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
01	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
01	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
00	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
01	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0

The decoder uses active-low outputs. Select

output = 0 (active) All unselected output = 1

(inactivate)

$$4-(4) \quad x = D_4 + D_5 + D_6 + D_7$$

$$y = D_2 + D_3 + D_6 + D_7$$

$$z = D_1 + D_3 + D_5 + D_7$$

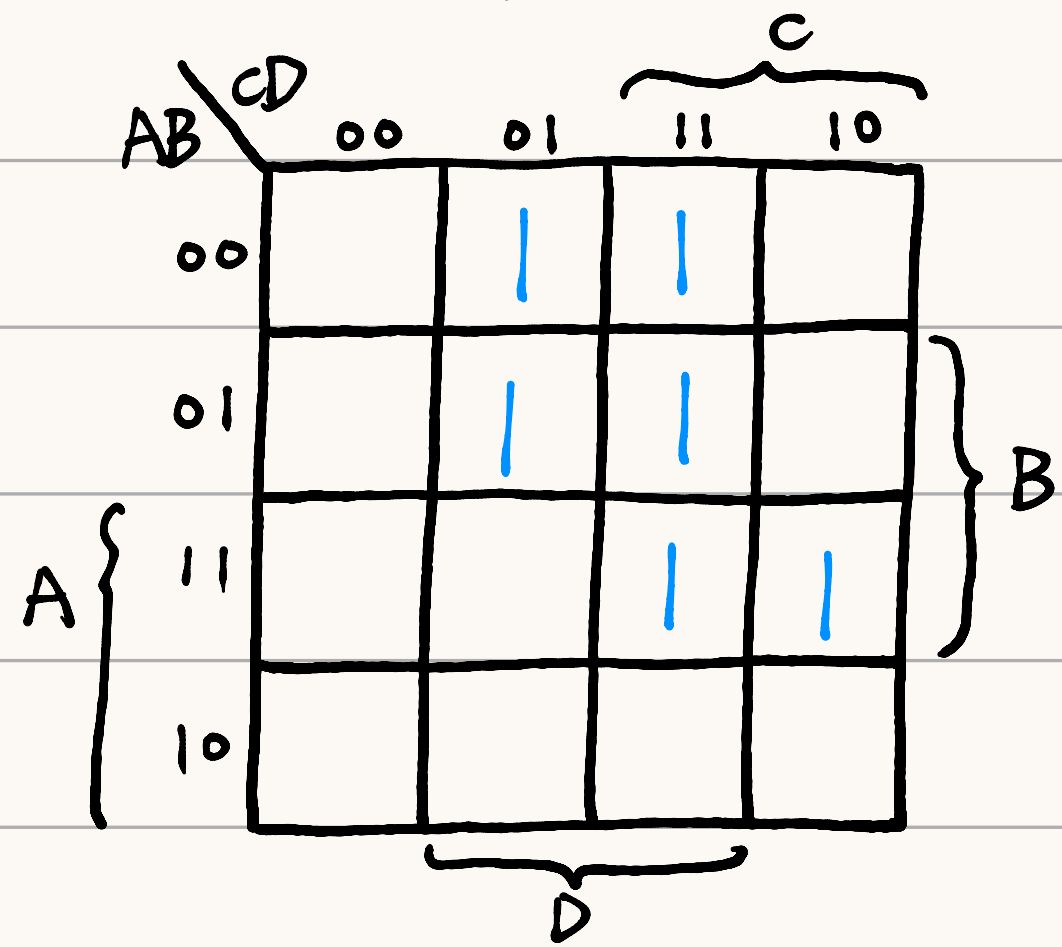
by truth table or k-map

Problem set :

$$4.2 \quad F = (A' (A'D')') (BC + A')$$

$$= (A + A'D)(BC + A')$$

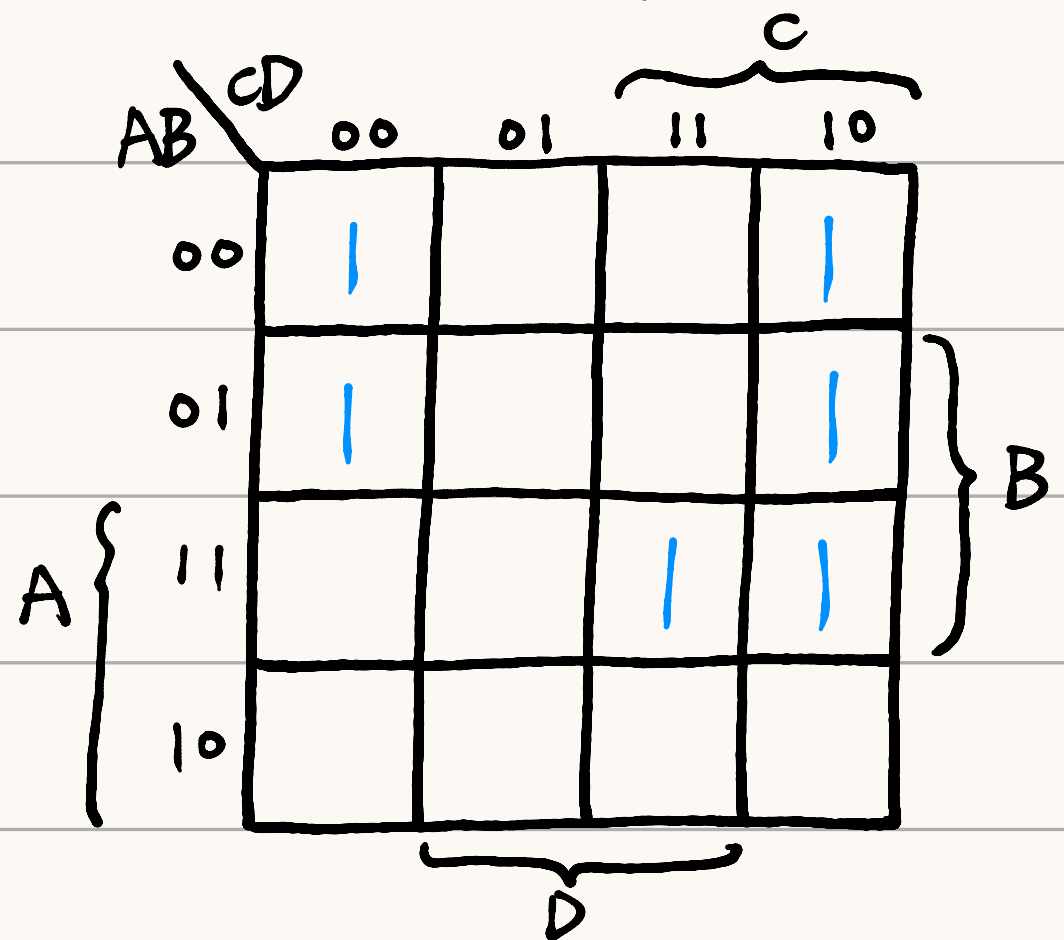
$$= ABC + A'BCD + A'D$$



$$F = A'D + ABC$$

$$G = (BC + A')(A'D')' = (A' + BC)(A + D')$$

$$= A'D' + ABC + BCD'$$



$$G = A'D' + ABC$$

4.4

x	y	z	F	F = z'
0	0	0	1	1
0	0	1	0	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	1	1
1	1	1	0	0

4.13 Sum C V

(a) 1101 0 1

(b) 0001 1 0

(c) 0100 1 0

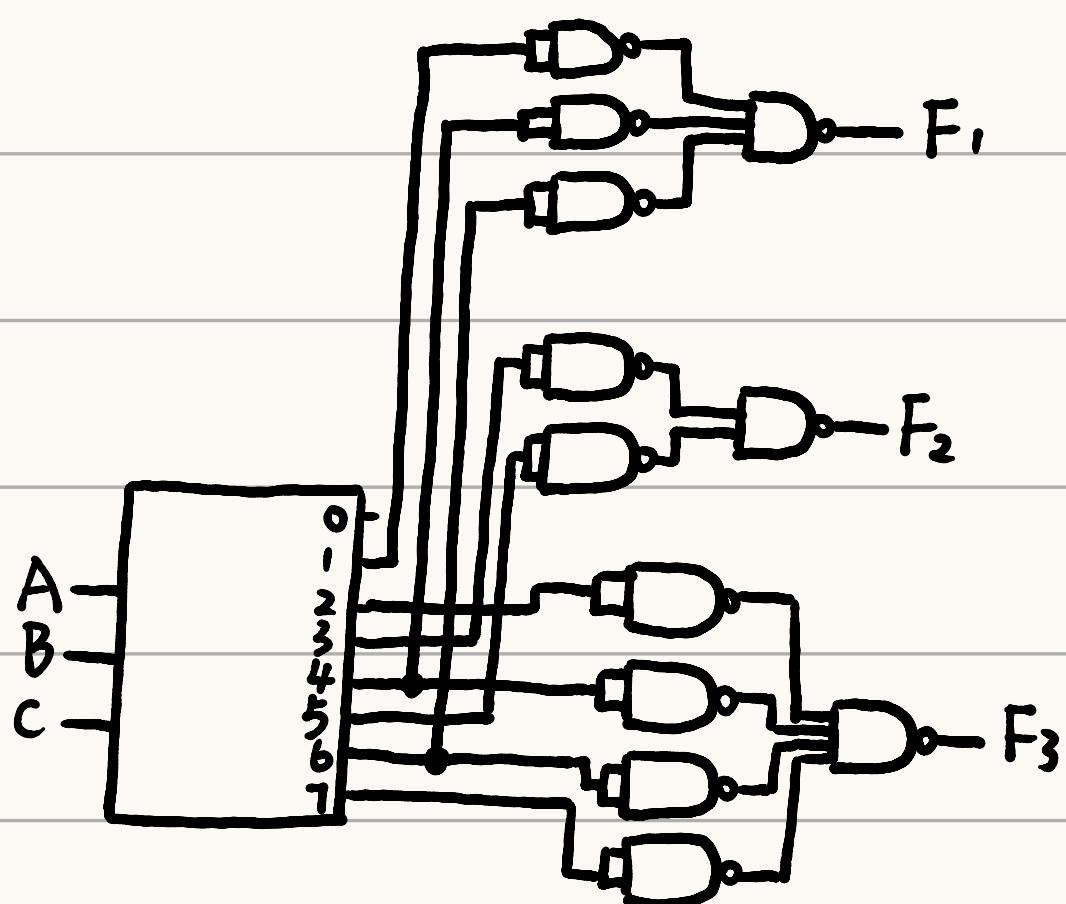
(d) 1011 0 1

(e) 1111 0 0

$$\begin{aligned}
 4.16 \quad (C_i' G_i' + P_i')' &= (C_i' (A_i B_i)' + (A_i + B_i)')' \\
 &= (C_i' (A_i B_i)')' (A_i + B_i) = (C_i + A_i B_i) (A_i + B_i) \\
 &= A_i C_i + B_i C_i + A_i B_i \\
 &= C_{i+1}
 \end{aligned}$$

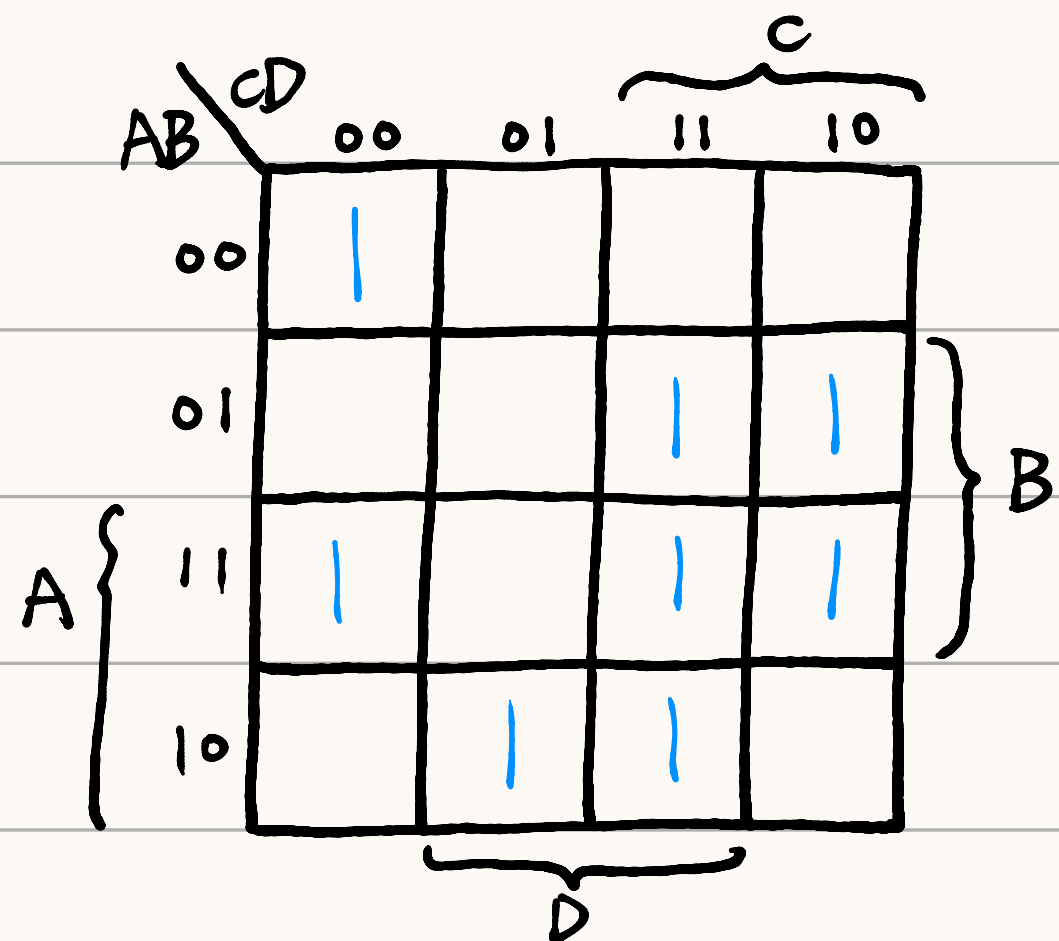
$$\begin{aligned}
 (P_i G_i') \oplus C_i &= [(A_i + B_i) (A_i B_i)'] \oplus C_i \\
 &= [(A_i + B_i) (A_i' + B_i')] \oplus C_i \\
 &= (A_i B_i' + A_i' B_i) \oplus C_i \\
 &= (A_i \oplus B_i) \oplus C_i \\
 &= S_{i+1}
 \end{aligned}$$

4.27



4.34

	A	B	C	D	F
I ₀	0	0	0	0	1
I ₁	0	0	1	0	0
I ₂	0	1	0	0	0
I ₃	0	1	1	0	1
I ₄	1	0	0	0	0
I ₅	1	0	1	0	0
I ₆	1	1	0	0	0
I ₇	1	1	1	0	1



$$F = \sum(0, 6, 7, 9, 11, 12, 14, 15)$$